

Dog Trick Competition 2 (dogtrick2)

RANDy is back at it again, but now he is entering a more advanced dog trick competition with his canine friend, Pompieru.

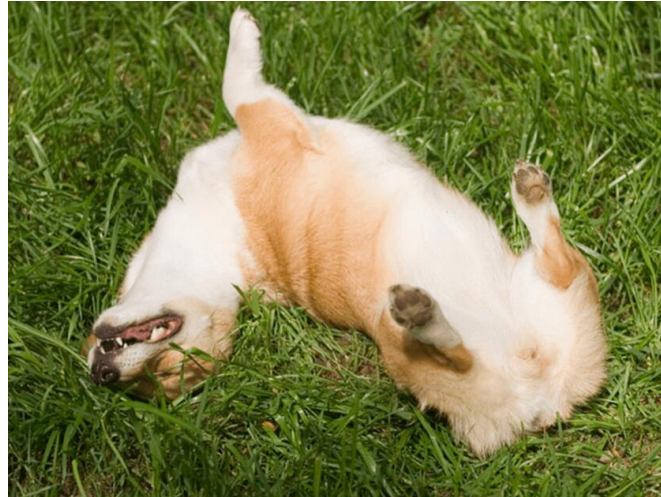


Figure 1: Pompieru training even more for the competition.

There are K tricks which can be performed in this competition, numbered from 1 to K and Pompieru knows how to perform all of them. However, performing specific tricks one after the other is not always possible: he can only perform trick b right after trick a if he has been trained to do so. RANDy trained Pompieru to perform M pairs of consecutive tricks (A_i, B_i) , meaning that Pompieru knows how to perform trick B_i right after trick A_i .

The participants are required to perform N tricks $T_0, T_1, \dots, T_{N-1}$ in this specific order. Pompieru will first perform the trick T_0 earning 2 points. When Pompieru finishes performing the trick T_i he will move on to trick T_{i+1} . If he can perform the trick T_{i+1} right after T_i he will earn 2 points, otherwise he can choose to perform some (one or more) in between tricks, enabling him to perform T_{i+1} . If he succeeds this way, he will earn 1 point. If not, he will be disqualified from the contest, earning as many points as he gathered prior to his disqualification.

Help RANDy and Pompieru find the number of points P they can score in this competition.

 Among the attachments of this task you may find a template file `dogtrick2.*` with a sample incomplete implementation.

Input

The input file consists of:

- a line containing integers N and K .
- a line containing the N integers $T_0, \dots, T_{N-1}$.
- a line containing integer M .
- M lines, the i -th of which consisting of integers A_i, B_i .

Output

The output file must contain a single line consisting of integer P .

Constraints

- $2 \leq N \leq 250\,000$.
- $1 \leq K \leq 250\,000$.
- $1 \leq M \leq 250\,000$.
- $1 \leq T_i \leq K$ for each $i = 0 \dots N - 1$.
- $T_i \neq T_{i+1}$ for each $i = 0 \dots N - 2$.
- $1 \leq A_i, B_i \leq K$ and $A_i \neq B_i$ for each $i = 0 \dots M - 1$.
- All the ordered pairs (A_i, B_i) are distinct.
- It is allowed to perform a trick multiple times.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (24 points) $N, K, M \leq 200$.

- **Subtask 3** (45 points) $1 \leq M \leq 100\,000, 1 \leq K \leq 50\,000$.

- **Subtask 4** (31 points) No additional limitations.


Examples

input	output
4 5 1 3 2 5 5 1 3 3 4 4 2 2 3 4 5	6
4 5 1 3 5 2 5 1 3 3 4 4 2 2 3 4 5	5

Explanation

In the **first sample case**, Pompieru earns 2 points for performing the first trick. He then earns 2 more points because he can perform trick 3 right after trick 1, then earns one more point performing trick 4 and then trick 2. After that he earns one more point by performing tricks 3, 4, and finally 5.

In the **second sample case**, Pompieru earns 2 points for performing the first trick. He then earns 2 more points because he can perform trick 3 right after trick 1, then earns one more point performing trick 4 and then trick 5. After that, there is no sequence of tricks he can do in order to perform trick 2, so he gets disqualified, earning a total of 5 points.